

MERN stack Technology

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What is a MERN Development Stack?

MERN is an acronym used to describe a specific set of JavaScript based technologies that are used in the web application development process. It is designed with an idea to make the development process as smooth as possible. MERN includes the following open-source components:

- Mongo DB
- Express JS
- React JS/ Redux
- Node JS



STACK



A “stack” refers to any combination of coding languages, tools, and frameworks developers use to create software products for their clients. A stack contains different layers of components that play a vital role in the process of software product creation.

- Client-side - front-end (HTML, CSS, JavaScript)
- Server-side - back-end (OS, web server, programming language, web framework, databases)



MongoDB



What is MongoDB?



This is a free open-source, cross-platform document-oriented database program. It is classified as a No SQL database program, which means that data is stored in flexible documents with JSON-based query language. This also means that the size of the content number of fields in the documents tends to vary. The whole data is structured in a way to be prone to change over time. MongoDB is known as a flexible solution that is always easy to scale.



Why MongoDB?



1. Schema-less: Data structure can vary per document.
2. High scalability with built-in sharding.
3. Rich querying and indexing capabilities.
4. Supports replication for fault tolerance.
5. Handles large volumes of data efficiently.

MongoDB Data Model

Data stored in documents (similar to JSON).

Documents grouped into collections (like tables in SQL).

Documents contain key-value pairs.

Example:

```
{  
  "name": "Ayan Ahmed",  
  "age": 20,  
  "skills": ["Python", "MongoDB", "Node.js"]  
}
```

MongoDB vs SQL Databases

Feature	MongoDB (NoSQL)	SQL Databases
Schema	Flexible, dynamic	Fixed, predefined schema
Data Storage	Document-based (BSON)	Table-based (rows/columns)
Scaling	Horizontal (sharding)	Vertical (scale-up)
Joins	Limited, aggregation pipeline	Full relational joins
Query Language	JSON-like queries	SQL

Basic CRUD Operations in MongoDB

Create: insertOne(), insertMany()

Read: find(), findOne()

Update: updateOne(), updateMany()

Delete: deleteOne(), deleteMany()

Example:

```
db.users.insertOne({ name: "Ayan", age: 22 });
```

```
db.users.find({ age: { $gt: 20 } });
```

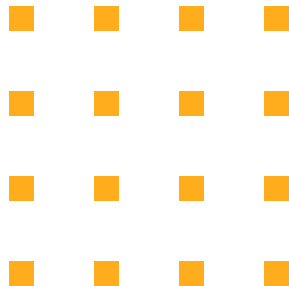
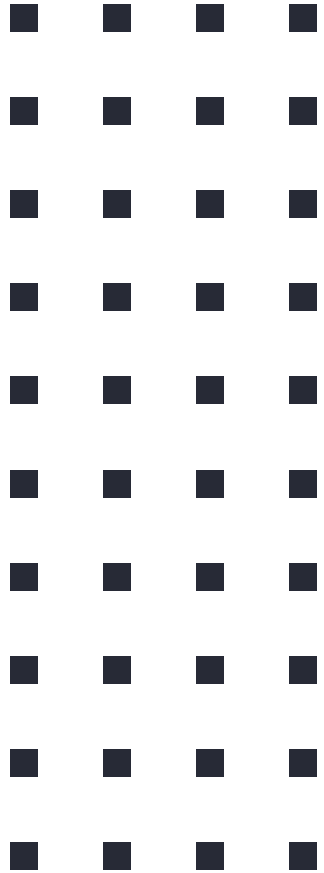


Use Cases of MongoDB



1. Real-time analytics and big data.
2. Content management systems.
3. Internet of Things (IoT) applications.
4. Mobile apps backend.
5. E-commerce catalogs.
6. Social networks and messaging platforms.

You can complete this free *Introduction to MongoDB* certification from the official MongoDB University: learn.mongodb.com/learning-paths/introduction-to-mongodb



Node.js & Express.js



What is **Node.js**?



Originally built for Google Chrome and later on open-sourced, Node JS is a cross-platform run-time JavaScript environment used for executing JavaScript code outside of a browser.

JavaScript was originally used for front-end scripting, but Node JS has enabled devs to use it to write command line tools and back-end scripts for the purpose of creating dynamic web page content before the page is sent to the user's web browser.

Node JS was designed with an idea of allowing devs to build scalable network applications.





Key Features

- **Asynchronous & Non-blocking I/O** – Handles multiple tasks without waiting for each to finish.
 - **Event-driven architecture** – Executes code in response to events for faster performance.
 - **Same Language for Frontend & Backend** – Developers can use JavaScript both on the client side and the server side, reducing context switching and speeding up development.
 - **Large NPM ecosystem** – Provides thousands of ready-to-use packages for rapid development.
 - **Cross-platform compatibility** – Runs seamlessly on Windows, macOS, and Linux.
- 

Your First Node.js App

```
Nodejs > JS workshop.js
```

```
1 console.log("hello, welcome to MERN workshop")
```

```
(base) PS C:\Users\garvi\OneDrive\Desktop\Nodejs> node "c:\Users\garvi\OneDrive\Desktop\Nodejs\workshop.js"  
hello, welcome to MERN workshop
```

Core Modules in Node.js

Node.js comes with built-in modules that handle essential backend tasks without extra packages.

1. **fs (File System)** – Read, write, update, and delete files.
2. **http** – Create web servers and handle HTTP requests/responses.
3. **path** – Work with file/directory paths in a platform-safe way.
4. **events** – Implement custom event-driven functionality.



Express JS



This is also a free, open-source software, it can be classified as a web application framework for Node.js. To be more precise, Express JS is made for developing web apps and APIs.

Instead of manually writing full web server code in Node.js, developers use this MERN component to simplify the coding process. The best feature of this framework is that devs don't repeat the same code over and over, as they would with writing Node.js code in the HTTP module.



Key Features



1. Lightweight and fast: Minimal overhead for quick server responses.
2. Middleware support: Allows processing requests in modular, reusable steps.
3. REST API ready: Designed to build RESTful endpoints efficiently.
4. Integration with MongoDB: Works seamlessly with MongoDB for database operations.

How Express.js Fits in the MERN Stack

1. **Frontend (React):** Users interact with React, which sends HTTP requests to the backend for data or actions.
2. **Backend (Express.js):** Express receives and processes these requests, applying routing and business logic.
3. **Database (MongoDB):** Express connects to MongoDB to perform data operations like create, read, update, and delete.
4. **Response Flow:** Express sends JSON responses back to React to update the user interface dynamically.



React.js

What is React? & Why use it?

This is a JavaScript library used for building user interfaces. Originally created by a software engineer who worked for Facebook, React was later on open-sourced.

Purpose: Helps create fast, interactive, and reusable UI components.

Key Features:

- Component-based architecture
- Virtual DOM for faster updates
- Unidirectional data flow (One-way binding) Strong community support

Real-world Use: Facebook, Instagram, Netflix, Airbnb, etc.



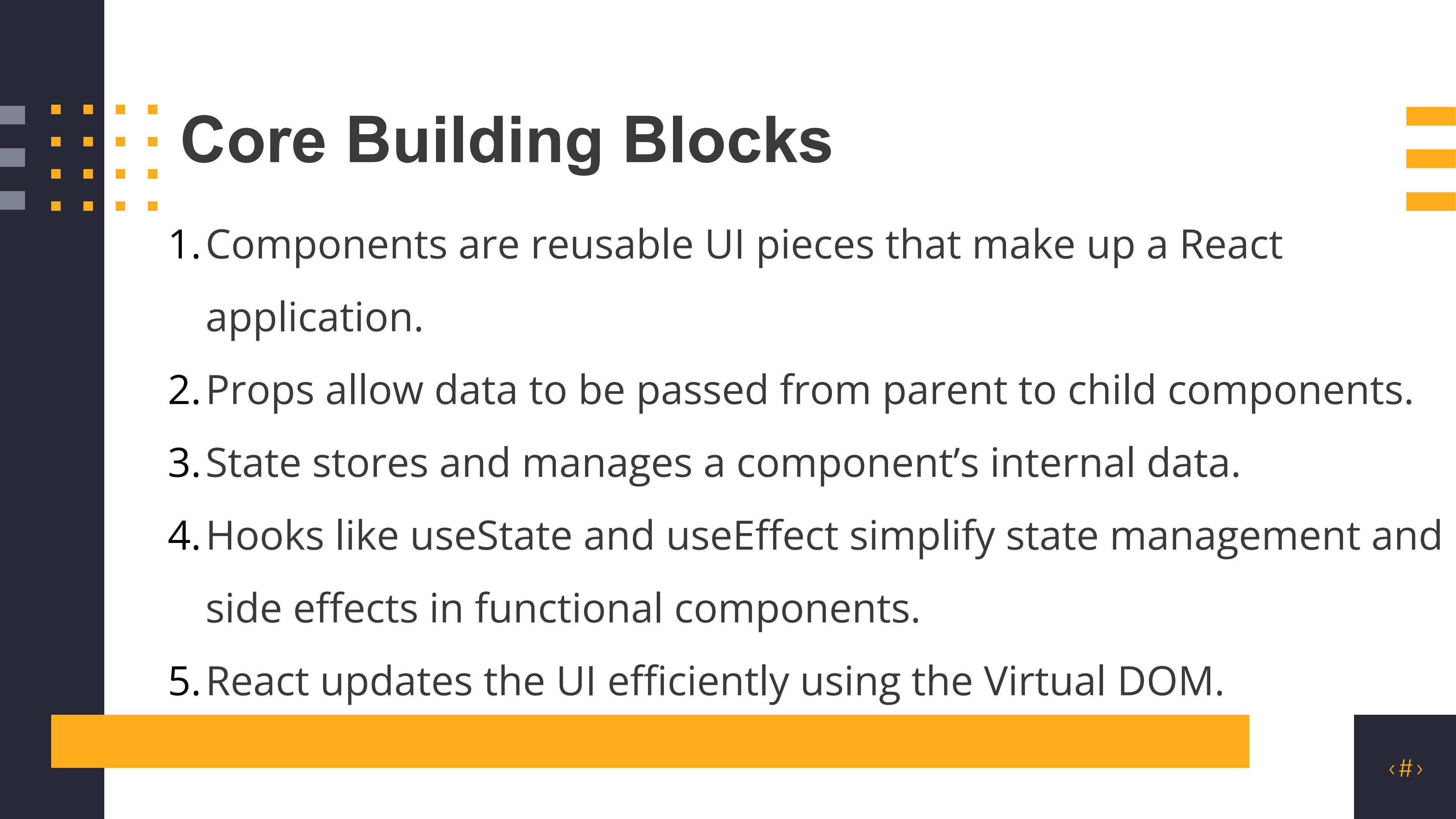
Getting Started with React



Install Node.js → Required for running React apps.
nodejs.org → Download & Install LTS version.

Create React App (CRA) Method:

```
npx create-react-app my-app  
cd my-app  
npm start
```



Core Building Blocks

1. Components are reusable UI pieces that make up a React application.
2. Props allow data to be passed from parent to child components.
3. State stores and manages a component's internal data.
4. Hooks like useState and useEffect simplify state management and side effects in functional components.
5. React updates the UI efficiently using the Virtual DOM.

Components, Props & State

Functional Component Example:

```
function Hello(props) {  
  return <h1>Hello, {props.name}</h1>;  
}  
export default Hello;
```

Using the Component:

```
import Hello from './Hello';  
function App() {  
  return <Hello name="John" />;  
}
```

Continued..

State with Hooks:

```
import { useState } from 'react';  
function Counter() {  
  const [count, setCount] = useState(0);  
  return <button onClick={() =>  
    setCount(count + 1)}>{count}</button>;  
}
```



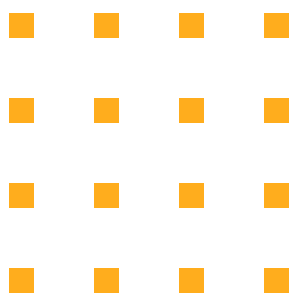
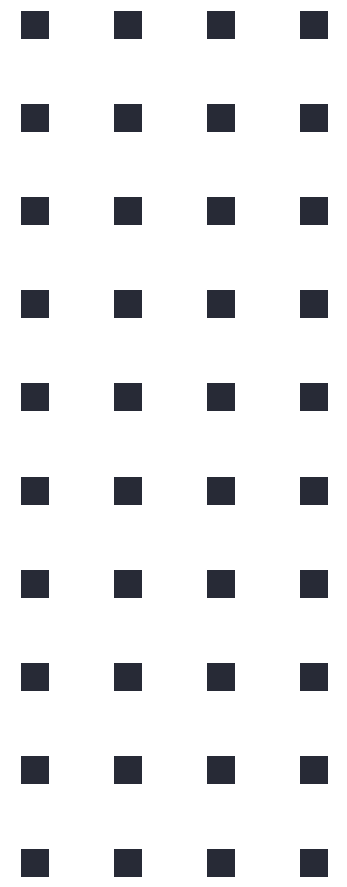
Advancing in React



1. Routing enables navigation between multiple pages without reloading the browser.
2. API Integration allows React apps to fetch and display live data from servers.
3. Performance Optimization involves techniques like memoization, lazy loading, and code splitting.
4. State Management Libraries (Redux, Zustand, Recoil) help manage complex app-wide state.
5. Following best practices—such as keeping components small and organized—makes apps scalable and maintainable.



MERN Stack in Internet of Things



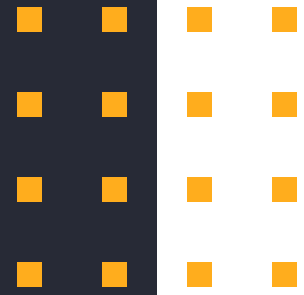


The Main Benefit of a MERN Stack

- JavaScript is everywhere. It is used both on the front-end and back-end side. Because of this, there's no need for context switching.
- MERN is JavaScript-based, devs only need to master a single coding language, which makes things a million times easier.
- building dynamic web applications in no time.

How MERN Stack is Used in IoT

- IoT Devices (ESP32, Raspberry Pi, Sensors) collect data → send via HTTP/WebSockets/MQTT.
- Node.js – Handles real-time, event-driven processing of device data.
- Express.js – Provides REST APIs/WebSocket endpoints for data transfer & device control.
- MongoDB – Stores flexible, JSON-like IoT data; supports large-scale time-series storage.
- React.js – Creates live dashboards for monitoring and controlling devices remotely.

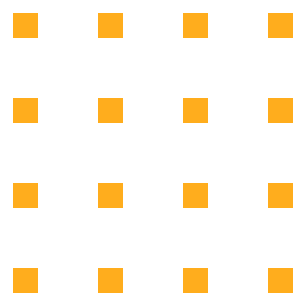
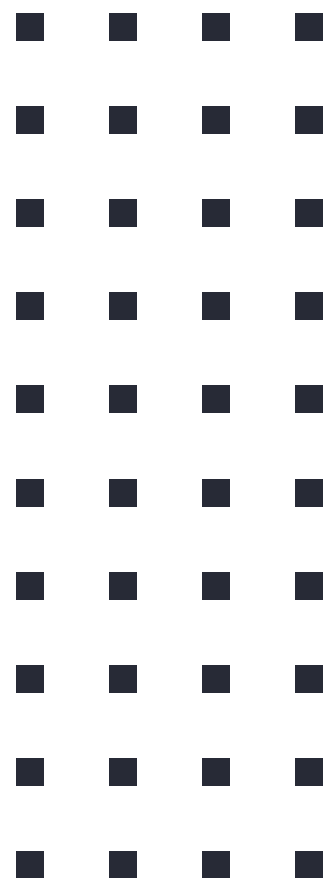


IoT Project Workshop

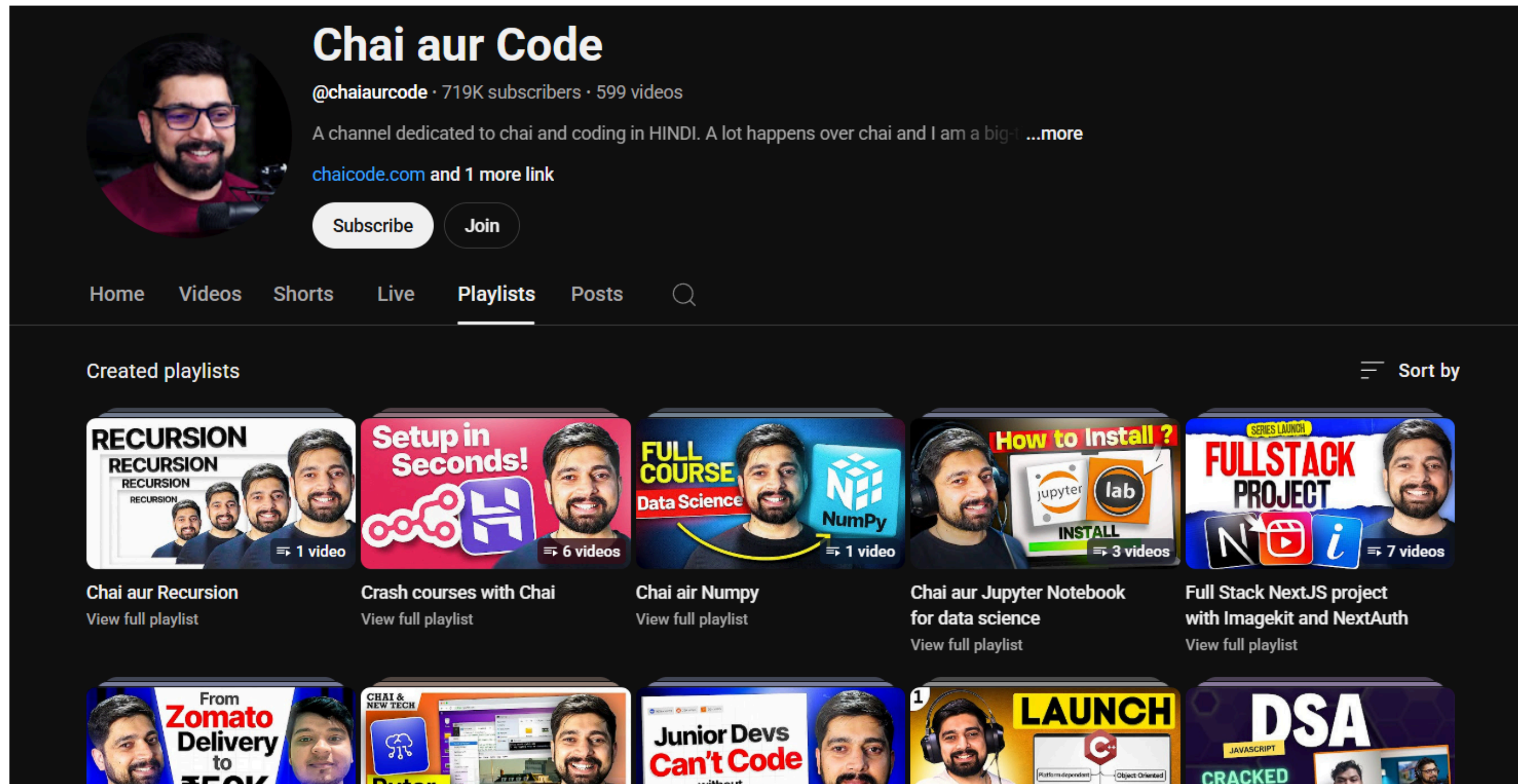
- <https://github.com/AyanAhmedKhan/Workshop-MernIoT>
- Star The Repository for future use or any open source contribution.
- Read carefully Readme file both Workshop_Guide.md and Readme.md file



Resources to start your journey



Chai aur Code – The Best for Industry-Level Learning: Simple, Clear & Easy to Understand



The screenshot shows the YouTube channel page for 'Chai aur Code'. The channel name is 'Chai aur Code' with the handle '@chaiaurcode', 719K subscribers, and 599 videos. The bio states: 'A channel dedicated to chai and coding in HINDI. A lot happens over chai and I am a big-t ...more'. There are links to 'chaicode.com' and '1 more link'. The 'Subscribe' button is highlighted. The navigation bar includes 'Home', 'Videos', 'Shorts', 'Live', 'Playlists' (which is selected), and 'Posts'. Below the navigation bar, the 'Created playlists' section is displayed with a 'Sort by' option. There are ten playlist thumbnails arranged in two rows of five. Each thumbnail features a video preview, a title, and a 'View full playlist' link.

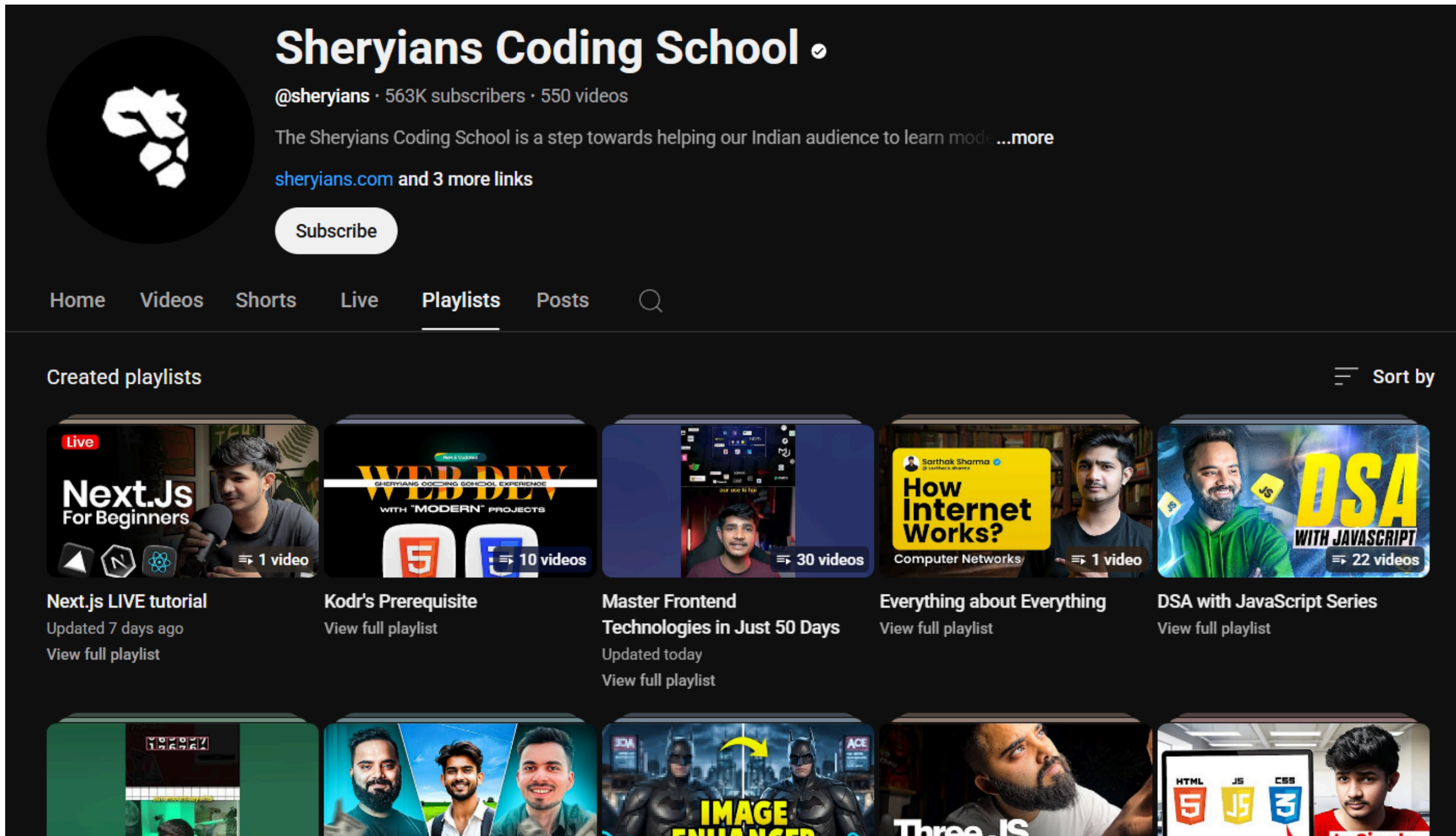
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- RECURSION**
Chai aur Recursion
View full playlist (1 video)
- Setup in Seconds!**
Crash courses with Chai
View full playlist (6 videos)
- FULL COURSE Data Science**
Chai air Numpy
View full playlist (1 video)
- How to Install?**
Chai aur Jupyter Notebook for data science
View full playlist (3 videos)
- FULLSTACK PROJECT**
Full Stack NextJS project with Imagekit and NextAuth
View full playlist (7 videos)
- From Zomato Delivery to 350K**
- CHAI & NEW TECH**
- Junior Devs Can't Code**
- LAUNCH**
- DSA CRACKED**

Sheryians Coding School - Best for Frontend



The screenshot shows the YouTube channel page for Sheryians Coding School. The channel name is "Sheryians Coding School" with a verified badge. The handle is @sheryians, with 563K subscribers and 550 videos. The bio states: "The Sheryians Coding School is a step towards helping our Indian audience to learn mode...more" and includes links to sheryians.com and 3 more links. A "Subscribe" button is visible. The navigation bar includes Home, Videos, Shorts, Live, Playlists (selected), and Posts. Under the "Created playlists" section, there are five featured playlists:

- Next.js LIVE tutorial**: Updated 7 days ago, View full playlist. (1 video)
- Kodr's Prerequisite**: View full playlist. (10 videos)
- Master Frontend Technologies in Just 50 Days**: Updated today, View full playlist. (30 videos)
- Everything about Everything**: View full playlist. (1 video)
- DSA with JavaScript Series**: View full playlist. (22 videos)

Below these, there are more playlist thumbnails visible, including "IMAGE ENHANCER" and "Three JS".

Code With Harry – Learn in a Structured & Proper Manner



Tutorial #1
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1



Tutorial #1
INSTALLING VS CODE & MORE

31:20

2



Tutorial #2
OUR FIRST HTML WEBSITE

28:31

3



Tutorial #3
BASIC STRUCTURE OF HTML FILE

11:12

4



Tutorial #4
HTML HEADING AND PARAGRAPHS

19:34

5



Tutorial #5
IMAGE, LISTS, AND TABLES IN HTML

19:34

Installing VS Code & How Websites Work | Sigma Web Development Course - Tutorial #1

CodeWithHarry · 5.6M views · 1 year ago

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donate.freecodecamp.org and 1 more link

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Front End Web Development
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- the freeCodeCamp.org podcast** (76 episodes)
Updated 4 days ago
View full podcast
- Back End Developer Roadmap** (18 lessons)
Back End Developer Learning Path
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Vue.js Live Coding with Gwen
- College Algebra** (10 videos)
with Python
FULL COURSE
- Intro to Python** (14 videos)
Python Basics with Sam

Bonus Free Certifications :

Bonus: Free Certifications & Job Simulations

1. Oracle – Race to Certification 2025

- Duration: July 1 to October 31, 2025
- Offer: Participating grants free digital training and 2 free exam attempts in your MyLearn wallet Oracle University+1.

2. Postman – API Fundamentals Student Expert Certification

- What you'll master: API basics, functional use of Postman, and real-world API interactions
- Certification: Free upon completion of the API Testing Path academy.postman.com+1.

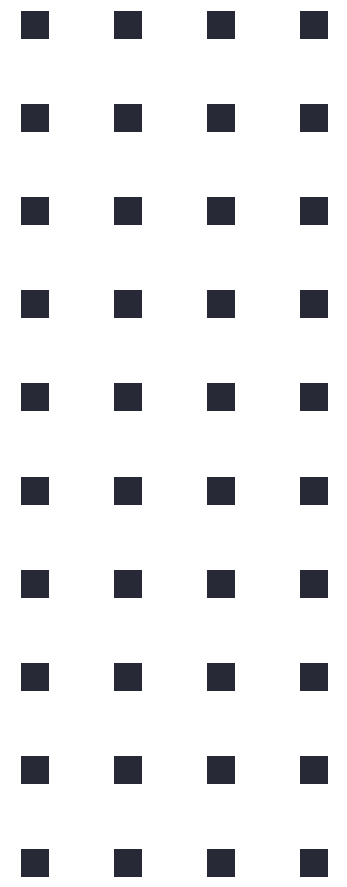
3. The Forage – Deloitte Australia: Data Analytics Job Simulation

- Format: Self-paced virtual internship (1–2 hours)
- Skills learned: Dashboard creation in Tableau, data analysis, data modeling, spreadsheet work
- Certificate: Earn a completion certificate to add to your resume or LinkedIn

By Ayan Ahmed Khan



THANKYOU



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